Carryover Effects and the Bias-Variance Tradeoff in Model Misspecification

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A White Paper on Non-Monotonic Power in Precision Medicine Trials

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Purpose. This paper, written in the pmsimstats2025 era (source `white_paper_carryover_bias_variance.md` of that project, now archived under `archive/pmsimstats2025-2026-04-23/`), examines how the power of a biomarker-treatment interaction test in a simplified open-label then discontinuation design varies with an unmodeled carryover half-life, and proposes a bias-variance explanation for the pattern.

Status note (2026-10-07). This is a historical document; a review on 2026-10-07 corrected its description of the simulation against the archived scripts, corrected its references, and found that several of its central claims are not supported by its own results. In brief: (i) the simulation is a single-path mean-moderation model with linear response growth, not the Hendrickson et al. (2020) model, and its "without carryover" analysis is misspecified even at $t_{1/2} = 0$; (ii) the simulated power does not show a clear drop-then-recovery pattern (Section 5); (iii) the derivation of Section 3.5 does not produce an interior maximum as originally stated; and (iv) the "precipitous decline" in the published Hendrickson results has since been traced to the step rule for the biomarker-response coupling under covariance moderation, not to a bias-variance tradeoff (`docs/36`, Sections 5.3 to 5.5; `docs/37`). Corrections are marked in place with the same date.

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Notation index and glossary

Symbols follow `analysis/report/NOTATION.md` where it defines one. Code names (`weeks_on_drug`, `bm_centered`) refer to the archived scripts and appear only in code font.

Table 1: Notation index for this paper

Symbol	Meaning	Status
Y_{it}	observed symptom score of participant i at week t	canonical
BR_{it}	biological response component	canonical
ε_{it}	observation-level residual, $N(0, 2^2)$ in the simulation	canonical
b_i	standardized biomarker (code `bm_centered`)	canonical
β_{bm}	mean-moderation multiplier, 0.3 in the simulation	canonical
$\beta_{bm:D}$	biomarker-by-exposure interaction coefficient of the analysis model; true value $-a\beta_{bm} = -0.15$	canonical
N	total number of participants (70; one path)	canonical
$t_{1/2}$	carryover half-life (weeks)	canonical
t_{od}, t_{sd}	cumulative weeks on drug (code `weeks_on_drug`); time since discontinuation	canonical
t_{od}^{disc}	weeks on drug at discontinuation (8 in the simulation)	local
$g(t_{sd})$	carryover retention, $2^{-t_{sd}/t_{1/2}}$ for $t_{1/2} > 0$, 0 for $t_{1/2} = 0$	local
a	response growth per week on drug (code `BR_rate`, 0.5)	local

Symbol	Meaning	Status
m, T	constant off-drug prediction of the analysis model; length of the off-drug span	local
$V(t_{1/2})$	integrated squared misspecification over the off-drug span	local
τ_1, $\tilde{\tau}_2$, ω_0, ω_1	period treatment effects and carryover effects in the crossover estimator of Section 2.3	local
c_{bm}	covariance-moderation correlation of the Hendrickson model (Section 5.5 only)	canonical
π	power, $\Pr(p < \alpha)$	canonical
α	nominal significance level, 0.05	canonical
n_{reps}	Monte Carlo replicates per cell (100)	canonical

Glossary.

- *OL+BDC*: open-label treatment followed by blinded discontinuation. In this paper's simulation every participant is on drug in weeks 1 to 8 and off drug thereafter; there is no randomized second path.
- *Carryover*: persistence of the drug effect after discontinuation, here exponential with half-life $t_{1/2}$.
- *Without / with carryover modeling*: analysis with exposure `weeks_on_drug` (constant after discontinuation) against analysis with an exponentially decayed exposure at an assumed half-life fixed at 0.2 weeks.
- *Bias %*: $100\,(\bar{\hat{\beta}}_{bm:D} - \beta_{bm:D})/|\beta_{bm:D}|$; since $\beta_{bm:D} < 0$, a positive value denotes attenuation toward zero.
- *MCSE*: Monte Carlo standard error, $\sqrt{\pi(1-\pi)/n_{\text{reps}}}$ for power.

Executive Summary

This white paper examines a counterintuitive phenomenon observed in simulations of Open-Label plus Blinded Discontinuation (OL+BDC) trial designs: when carryover effects are not modeled in the analysis, statistical power exhibits non-monotonic behavior as carryover half-life increases. Specifically, power

drops sharply from zero carryover to short carryover ($t_{1/2}$ = 0.1-0.5 weeks), then partially recovers as carryover extends further ($t_{1/2}$ = 1-2 weeks).

This phenomenon reflects a fundamental bias-variance tradeoff arising from model misspecification and has important implications for trial design and analysis.

Correction (2026-10-07). The paper's own simulation (Section 5.1) does not reproduce this pattern clearly. Without carryover modeling, power is 0.06 at $t_{1/2} = 0$, 0.00-0.02 at 0.05-0.5 weeks and rises to 0.36 at 2 weeks (simulated, 100 replicates per cell; MCSE up to 0.05). The drop from 0.06 to 0.01 is about two MCSEs of a difference, and at $t_{1/2} = 0$ the mean estimate has the wrong sign, so the evidence is for a power that increases with half-life beyond about 0.2 weeks rather than for a drop followed by recovery. The pattern quoted in this summary comes from the unsourced table of Section 1.1.

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1. Introduction

1.1 The Observed Phenomenon

In simulations comparing trial designs for precision medicine validation (Hendrickson et al., 2020), we observed the following pattern for OL+BDC designs analyzed WITHOUT explicit carryover modeling:

Table 2: OL+BDC power by carryover half-life without explicit carryover modeling (approximate values; source not recorded)

Carryover $t_{1/2}$	Statistical Power
0 weeks	~85% (highest)
0.1 weeks	~60% (sharp drop)
0.2 weeks	~55% (near minimum)
0.5 weeks	~58% (slight recovery)
1.0 weeks	~62% (continued recovery)
2.0 weeks	~68% (partial recovery)

Note (2026-10-07). The source of these approximate values is not recorded, and a search of the archived pmsimstats2025 analysis scripts for this half-life grid found none; they are unverified. The half-life grid matches neither the Hendrickson et al. (2020) grid (0, 0.1, 0.2 weeks) nor exactly the grid of Section 5, and the baseline of about 85% differs from the 6% simulated at $t_{1/2} = 0$ in Section 5.1.

This non-monotonic pattern contradicts the naive expectation that increasing carryover should monotonically degrade power when unmodeled.

1.2 Significance

Hendrickson et al. (2020) noted that "the presence of even a short (0.1 weeks) carryover component resulted in a **precipitous decline in power**" in both N-of-1 and OL+BDC designs (the full sentence reads "in both the N-of-1 and, to a lesser but still very significant degree, the open label + blinded discontinuation design"; quotation verified against the PubMed Central full text). The magnitude of this drop underscores the importance of carryover as a design parameter. The published grid stops at 0.2 weeks, so any recovery at longer half-lives lies outside it.

Understanding this phenomenon is essential for:

1. Appropriate trial design selection
2. Deciding whether to model carryover in the analysis
3. Setting realistic power expectations across carryover scenarios

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2. Background: Carryover in Crossover and N-of-1 Designs

2.1 Definition of Carryover

Carryover (or residual) effects occur when the pharmacological or biological effect of a treatment persists into subsequent periods after treatment discontinuation. In the context of our simulations, carryover follows an exponential decay model:

$$g(t_{sd}) = \left(\tfrac{1}{2}\right)^{t_{sd}/t_{1/2}},$$

where t_{sd} is time since discontinuation and $t_{1/2}$ is the half-life; $g = 0$ off drug when $t_{1/2} = 0$.

2.2 Literature Guidance

The crossover trial literature provides extensive guidance on carryover (Wellek & Blettner, 2012; Lim & In, 2021):

- **Washout periods**: Standard practice sets washout at 3-5× the pharmacokinetic half-life (attributed; not checked against the full texts of the two reviews)
- **Statistical tests**: Grizzle's (1965) two-stage procedure tests for carryover before analyzing treatment effects, though Freeman (1989) showed that it inflates the Type I error and judged it too misleading for practical use

- **Model-based approaches**: Mixed effects models can incorporate carryover terms, but are sensitive to misspecification

Recent work on behavioral carryover (Shi & Ye, 2024) distinguishes between:

- **Biological carryover**: Pharmacokinetic persistence (addressable via washout)
- **Behavioral carryover**: Treatment-induced behavior changes that persist regardless of washout

2.3 Mathematical Framework in the Literature

Shi and Ye (2024) formalize carryover in potential outcomes notation (their symbols are renamed here to avoid collision with the project's canonical λ and θ):

- ω_0: carryover effect when the previous treatment was control
- ω_1: carryover effect when the previous treatment was active

With τ_1 and $\tilde{\tau}_2$ the treatment effects in the two periods, the basic crossover estimator $\hat{\tau}$ has expectation

$$E(\hat{\tau}) = \tfrac{1}{2}(\tau_1 + \tilde{\tau}_2 - \omega_0 - \omega_1).$$

This shows the estimator is biased by the combined carryover term $\omega_0 + \omega_1$. The direction and magnitude of bias depend on the sign and size of this sum. (Unverified in detail: the published abstract confirms that under a sign condition on the carryover effect the basic estimator underestimates the treatment effect, but this expression was not checked against the full text.)

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3. Mathematical Framework for the Non-Monotonic Phenomenon

3.1 The OL+BDC Design Structure

In an OL+BDC design:

- **Open-label phase** (weeks 1-8): All participants receive active treatment
- **Blinded discontinuation** (weeks 9+): Participants randomized to continue or discontinue

Correction (2026-10-07, inspected `simulation_carryover_spectrum.R`). The simulation does not randomize at discontinuation: every participant is on drug in weeks 1 to 8 and off drug from week 9, with measurements at weeks 1 to 12, 16 and 20. It is therefore a single-path design with no concurrent on-drug comparison during the discontinuation phase.

The analysis predictor in a simple model (without carryover) is `weeks_on_drug`, which accumulates during treatment but remains constant after discontinuation.

3.2 True Data-Generating Process

Let BR_{it} denote the true biological response of participant i at week t. *Correction (2026-10-07).* The original text called this "the Hendrickson model", used a Gompertz growth function, and applied the moderation factor twice off drug. The archived script (`simulation_carryover_spectrum.R`, inspected) instead uses a mean-moderation model with linear growth, which differs from the covariance-moderation model of Hendrickson et al. (2020):

During treatment:

$$BR_{it} = a\, t_{od,it}\, (1 + \beta_{bm} b_i),$$

After discontinuation ($t_{1/2} > 0$):

$$BR_{it} = a\, t_{od}^{\text{disc}}\, (1 + \beta_{bm} b_i)\, g(t_{sd,it}),$$

and $BR_{it} = 0$ off drug when $t_{1/2} = 0$. The outcome is $Y_{it} = 50 - BR_{it} + \varepsilon_{it}$, with no random intercept or serial correlation in the data-generating process, where:

- $a = 0.5$ is the response growth per week on drug
- $\beta_{bm} = 0.3$ is the biomarker moderation multiplier (the parameter of interest), applied to the standardized biomarker b_i
- $t_{od}^{\text{disc}} = 8$ is the number of weeks on drug at discontinuation
- t_{sd} is time since discontinuation, counted from the first off-drug visit (week 9, where $t_{sd} = 0$ and $g = 1$ for any $t_{1/2} > 0$)
- $t_{1/2}$ is the carryover half-life
- $\varepsilon_{it} \sim N(0, 2^2)$ independently

3.3 Analysis Model (Without Carryover)

The analysis model assumes:

```
symptoms ~ biomarker + weeks_on_drug + biomarker:weeks_on_drug + (1|participant)
```

The key interaction term `biomarker:weeks_on_drug` is used to detect biomarker moderation of treatment response. In the script the model is fitted with `lme4::lmer` by maximum likelihood, with the standardized biomarker `bm_centered` in place of `biomarker`, and the interaction is tested with a t reference distribution on $N - 4 = 66$ degrees of freedom. Its target is $\beta_{bm:D} = -a\beta_{bm} = -0.15$, the on-drug slope. The "with carryover" analysis of Section 5 replaces `weeks_on_drug` off drug by $t_{od}^{\text{disc}}\, 2^{-t_{sd}/0.2}$, that is, it assumes a half-life of 0.2 weeks whatever the true value.

3.4 Source of the Bias-Variance Tradeoff

The non-monotonic power arises from competing effects:

Component 1: Signal Attenuation (Monotonically Increasing with $t_{1/2}$) As $t_{1/2}$ increases, the true effect during off-drug periods increasingly differs from zero. But the analysis predictor (`weeks_on_drug`) remains constant at its discontinuation value. This creates:

- **At $t_{1/2} = 0$**: Perfect on/off separation. Off-drug has zero true effect and predictor is irrelevant for those periods.
- **At $t_{1/2} > 0$**: The "off-drug" periods have residual effect that the predictor cannot capture.

Correction (2026-10-07, derived from the model of Section 3.2). The first bullet does not hold for the predictor actually used. Because `weeks_on_drug` stays at 8 after discontinuation, at $t_{1/2} = 0$ the model predicts the full on-drug interaction at every off-drug visit while the true effect is zero: this is the largest mismatch between predictor and truth, not perfect separation. The mismatch shrinks monotonically as $t_{1/2}$ grows and the true off-drug effect approaches the constant predictor. This is consistent with the simulated estimates (Section 5.1), which are attenuated by 92-96% at short half-lives and less at long ones, and with the wrong-signed mean estimate at $t_{1/2} = 0$.

Component 2: Unexplained Variance (Non-Monotonic with $t_{1/2}$) The original argument turns on how unexplained variance in off-drug observations varies with $t_{1/2}$:

At $t_{1/2} \approx$ 0.1-0.5 weeks (short carryover): - True effect decays rapidly during off-drug (at $t_{1/2} = 0.1$, from 100% at the first off-drug visit to 0.1% one week later; corrected 2026-10-07 from "12% in 3 weeks", which is the $t_{1/2} = 1$ value) - This creates high WITHIN-PERIOD variance in response - The predictor is constant, but the outcome varies substantially - Maximum mismatch → Maximum residual variance → **Minimum power**

At $t_{1/2} \approx$ 2+ weeks (long carryover): - True effect decays slowly (at $t_{1/2} = 2$, to 71% after one week and 35% after three; corrected 2026-10-07 from "70% in 3 weeks") - Off-drug response remains relatively stable - The predictor is constant, AND the outcome is roughly constant - Reduced mismatch → Reduced residual variance → **Power recovery**

3.5 Formal Expression

The unexplained variance in the model residuals can be described heuristically as the sum of within-participant noise, between-participant variation, and a misspecification term, the mean squared difference between the true response and the model's prediction. For off-drug observations the true response is $A\, g(t_{sd})$, with $A = a\, t_{od}^{\mathrm{disc}}(1 + \beta_{bm} b_i)$ the response at discontinuation, while the analysis model, whose predictor is constant off drug, predicts a constant m. Over an off-drug span of length T the integrated squared misspecification is

$$V(t_{1/2}) = \int_0^T \left[A\, 2^{-s/t_{1/2}} - m\right]^2 ds.$$

Correction (2026-10-07, derived). The original text stated that this integral has an interior maximum in $t_{1/2}$. Whether it does depends on m:

- m **fixed at the on-drug level,** $m = A$**.** This is what a constant `weeks_on_drug` predictor with a common slope implies. For every s, $2^{-s/t_{1/2}}$ increases with $t_{1/2}$, so $(A\, 2^{-s/t_{1/2}} - A)^2$ decreases, and V falls monotonically from $A^2 T$ as $t_{1/2} \to 0$ to 0 as $t_{1/2} \to \infty$. There is no interior maximum.
- m **the least-squares constant over the span.** Then $V = T \operatorname{Var}_s(A\, 2^{-s/t_{1/2}})$, which is 0 in both limits (the trajectory is constant at 0 or at A) and positive in between, so an interior maximum exists.

(The original limits were also misstated: at $t_{1/2} \to 0$ the integral is $m^2 T$, not m^2.) The fitted model is neither case exactly, because one interaction slope serves both phases, but the simulated estimates of Section 5.1, whose attenuation is largest at $t_{1/2} = 0$ (where the estimate changes sign), roughly flat at 95% up to 0.2 weeks and shrinking thereafter, behave like the first case. The bias-variance account of a minimum at intermediate half-life is therefore not supported by this derivation.

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4. Intuitive Explanation

Consider an analogy: imagine trying to predict a person's location using only their home address as a predictor.

Scenario A: Person stays at home ($t_{1/2} = 0$)

- Predictor: Home address
- True location: Always at home
- Result: **Perfect prediction**

Scenario B: Person commutes daily ($t_{1/2} = 0.5$)

- Predictor: Home address
- True location: Varies between home, transit, and work
- Result: **Poor prediction**: high unexplained variance

Scenario C: Person works from home ($t_{1/2} = \infty$)

- Predictor: Home address
- True location: Almost always at home
- Result: **Good prediction**: effect persists, matching the constant predictor

The intermediate case (Scenario B) creates the most unexplained variance because the true outcome varies while the predictor remains constant.

Note (2026-10-07). The analogy follows the original hypothesis. For the constant `weeks_on_drug` predictor actually simulated, $t_{1/2} = 0$ is the case of largest mismatch (Section 3.4), so Scenario A does not describe the simulated analysis.

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5. Simulation Evidence

5.1 Fine-Grained Carryover Spectrum Simulation

We conducted a simulation study across 11 carryover half-life values (0, 0.05, 0.1, 0.15, 0.2, 0.3, 0.5, 0.75, 1.0, 1.5, 2.0 weeks) with 100 iterations each (simulated; $N = 70$, seed 42). The results show a large effect of carryover modeling. All values below were re-read from the archived `carryover_spectrum_results.RData` on 2026-10-07 and agree with the table (verified); no fit failed.

Table 3: Power and bias by carryover half-life with and without carryover modeling (source: simulation_carryover_spectrum.R, N = 70, 100 iterations; the with-carryover analysis assumes a half-life of 0.2 weeks; positive bias denotes attenuation toward zero)

$t_{1/2}$ (weeks)	Power (Without)	Power (With)	Bias % (Without)	Bias % (With)
0.00	6%	100%	+113%	+28%
0.05	1%	100%	+95%	-2%
0.10	1%	100%	+96%	-1%
0.15	2%	100%	+95%	-1%
0.20	0%	100%	+96%	-1%
0.30	2%	100%	+92%	+4%
0.50	2%	100%	+88%	+3%
0.75	8%	100%	+84%	+12%
1.00	12%	100%	+80%	+16%
1.50	17%	100%	+73%	+24%
2.00	36%	100%	+64%	+32%

Power is a proportion of 100 replicates, so its MCSE is at most 0.05 (0.048 at 0.36, 0.024 at 0.06, 0 at 100%). The bias is relative to the on-drug target $\beta_{bm:D} = -0.15$; a bias above +100%, as at $t_{1/2} = 0$ without carryover modeling (mean estimate +0.020), means the estimate has the wrong sign.

5.2 Key Findings from Simulation

Without Carryover Modeling:

1. **Severe bias at short carryover**: At $t_{1/2} = 0.05$-0.3 weeks, estimates are biased toward zero by 92-96%, making the biomarker moderation effect nearly undetectable.
2. **Power increases with longer carryover**: Counter to naive expectation, power INCREASES from 0-2% at short carryover to 36% at $t_{1/2} = 2$ weeks.
3. **Mechanism**: Longer carryover means the off-drug periods retain more of the treatment effect. Since the analysis predictor (`weeks_on_drug`) remains constant after discontinuation, it becomes a better proxy for the true cumulative effect when carryover persists.

With Carryover Modeling:

1. **Uniform high power**: The carryover-adjusted analysis rejects in 100 of 100 replicates in every carryover scenario.

2. **Moderate bias**: Bias ranges from -2% to +32%. It is smallest where the true half-life is near the assumed 0.2 weeks (-2% to -1% at 0.05-0.2) and grows as the true half-life departs from it, in either direction (+28% at 0, +32% at 2 weeks). It is therefore a half-life misspecification bias of the adjusted analysis (corrected 2026-10-07 from "a separate issue related to model parameterization").

5.3 Predictor-Outcome Correlation Analysis

Analysis from `diagnose_nonmonotonic_power.R` shows how the correlation between the simple predictor (`weeks_on_drug × biomarker`) and the true biological response varies with carryover:

Table 4: Correlation between the simple predictor (weeks on drug times standardized biomarker) and true BR, and its square, by half-life (one simulated data set per half-life)

Carryover $t_{1/2}$	Cor(predictor, true_BR)	Squared correlation
0.0 weeks	0.214	0.046
0.5 weeks	0.342	0.117
1.0 weeks	0.421	0.177
2.0 weeks	0.525	0.276

Correction (2026-10-07, verified by rerunning `diagnose_nonmonotonic_power.R`). The correlations reproduce exactly. The third column was labeled R^2 but is the square of the second column; the script's own R^2 of `weeks_on_drug` alone against true BR is 0.084, 0.013, 0.004 and 0.000, which falls with half-life. Each row is a single data set of 70 participants, the diagnostic uses $\beta_{bm} = 0.5$ rather than 0.3 and measurement weeks 4, 8, 9-12, 16 and 20, so it is a descriptive illustration, not a replicated result.

Counterintuitively, the correlation INCREASES with longer carryover. This is because:

1. At $t_{1/2} = 0$, off-drug observations have zero true effect, contributing zero covariance but positive predictor variance
2. At $t_{1/2} > 0$, off-drug observations have positive true effect that correlates with the (constant) predictor value

5.4 Off-Drug Retention Profiles

The carryover retention at off-drug timepoints (the last dose is in week 8; t_{sd} is counted from the first off-drug visit, week 9):

Table 5: Carryover retention at off-drug weeks by carryover half-life (values of $2^{-t_{sd}/t_{1/2}}$, checked 2026-10-07)

Week	t_{sd}	$t_{1/2} = 0.1$	$t_{1/2} = 0.2$	$t_{1/2} = 0.5$	$t_{1/2} = 1.0$	$t_{1/2} = 2.0$
9	0	1.000	1.000	1.000	1.000	1.000
10	1	0.001	0.031	0.250	0.500	0.707
12	3	0.000	0.000	0.016	0.125	0.354
16	7	0.000	0.000	0.000	0.008	0.088
20	11	0.000	0.000	0.000	0.000	0.022

At $t_{1/2} = 0.1$-0.2 weeks, the effect has essentially vanished by week 10, creating a mismatch between the constant predictor and the rapidly decaying true effect.

At $t_{1/2} = 2$ weeks, substantial effect (35-70%) persists through week 12, making the off-drug period appear more similar to the on-drug period, reducing the model misspecification.

5.5 Reconciling with Hendrickson et al. Findings

Hendrickson et al. (2020) reported that "even a short (0.1 weeks) carryover component resulted in a precipitous decline in power." Our simulation confirms this finding:

- Power drops from ~6% at $t_{1/2} = 0$ to 0-2% at $t_{1/2} = 0.1$-0.3 weeks
- This represents a collapse in detection ability

Correction (2026-10-07). This simulation does not confirm the published finding. The drop from 6% to 0-2% is within about two MCSEs, starts from a baseline that is already near the nominal level, and at $t_{1/2} = 0$ the mean estimate has the wrong sign; there is no collapse from high power to reproduce. The published collapse (Hybrid power at $N = 70$, $c_{bm} = 0.3$ falling from 0.74 to 0.12 at $t_{1/2} = 0.1$) arises in a covariance-moderation model and has since been explained by the published step rule for the biomarker-response coupling: with any positive half-life the coupling is constant across visits, the coupling gap between on-drug and off-drug visits is zero, and the interaction vanishes; graded coupling ($2^{-t_{sd}/t_{1/2}}$ off drug, 0 before first exposure) removes the collapse (`docs/36`, Sections 5.3 to 5.5 and 6.8; `docs/37`). A mean-moderation model with linear growth, as here, cannot exhibit that mechanism.

The apparent non-monotonicity observed in the original heatmaps (power dropping then recovering) depends on the specific design parameters and the baseline power at $t_{1/2} = 0$. When baseline power is already low (as in our simplified simulation), the pattern appears as monotonic recovery with longer carryover. When baseline power is high (as in the full Hendrickson simulation with more complex

data generation), the drop-then-recovery pattern is more pronounced. (This explanation was not tested: no high-baseline run of the simplified model was made, and the published Hendrickson grid does not extend beyond 0.2 weeks.)

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6. Implications and Recommendations

6.1 Trial Design Implications

1. **Very short carryover ($t_{1/2}$ < 0.2 weeks)** represents the "worst case" for analysis without carryover modeling, not zero carryover.
2. **Zero carryover** is actually the **best case** for simple analysis models, providing clean on/off separation.

Note (2026-10-07). Items 1 and 2 restate the original hypothesis. In this paper's simulation power without carryover modeling is 0.00-0.06 over 0 to 0.5 weeks with no reliable difference between zero and very short carryover (Section 5.1), and for the constant `weeks_on_drug` predictor zero carryover is the case of largest mismatch (Section 3.4).

3. **Long carryover ($t_{1/2}$ > 1 week)** partially recovers power even without explicit carryover modeling, because the effect remains relatively stable.

6.2 Analysis Recommendations

1. **When carryover is suspected but unknown:**
   - Always include carryover modeling in the analysis
   - The exponential decay model is robust across different true decay forms (exponential, linear, Weibull) per our robustness simulations (not reported or cited in this paper; unverified here)
2. **Power calculations:**
   - Do not assume power degrades monotonically with carryover
   - The minimum power scenario is at intermediate carryover ($t_{1/2} \approx$ 0.1-0.5 weeks for typical trial durations; in this paper's simulation the minimum is not distinguishable from the $t_{1/2} = 0$ value)
3. **Washout periods:**
   - If carryover modeling is not feasible, longer washout periods should target at least 5× the expected half-life
   - For behavioral or unknown carryover, analytical approaches (modeling or down-weighting early post-discontinuation observations) are preferable

6.3 Future Research Directions

1. Optimal down-weighting schemes for early post-discontinuation observations
2. Robust estimation methods that are insensitive to carryover misspecification
3. Adaptive designs that can estimate carryover parameters during the trial

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7. Conclusions

The relationship between carryover half-life and statistical power in simple analysis models reflects a fundamental bias-variance tradeoff arising from model misspecification. Our key findings are:

7.1 Primary Conclusions

1. **Carryover modeling is essential**: Without explicit carryover modeling, power ranges from 0-36% across the carryover spectrum. With carryover modeling, power is uniformly 100% in this simplified simulation. The difference shows the importance of appropriate model specification.

2. **Short carryover is the worst case**: At $t_{1/2} = 0.1$-0.3 weeks, the simple model exhibits 92-96% bias toward zero, making effect detection nearly impossible (0-2% power).

3. **Longer carryover paradoxically helps simple models**: As carryover increases beyond 0.5 weeks, power gradually recovers (up to 36% at $t_{1/2} = 2$ weeks) because the persistent effect better matches the constant predictor assumption.

4. **The "precipitous decline" is real**: Hendrickson et al.'s observation that even 0.1 weeks of carryover causes dramatic power loss is confirmed. This represents the critical transition from a well-specified to a misspecified model.

Correction (2026-10-07). Conclusion 4 is withdrawn. The simulation starts from a power of 0.06, not from a well-specified high-power model, so it cannot confirm a collapse; and the published decline has been traced to the step coupling rule of the covariance-moderation model (`docs/36`, Section 5; `docs/37`), not to the misspecification studied here. Conclusion 2 should also be read with Section 3.4: the worst case for the constant predictor is $t_{1/2} = 0$ in terms of bias, and power at 0 and at 0.1-0.3 weeks is not reliably different.

7.2 Practical Recommendations

1. **Always model carryover** when any carryover is suspected, regardless of the expected half-life.
2. **Exponential decay is robust**: Our robustness analysis shows the exponential carryover model performs well even when the true decay follows linear or Weibull patterns.
3. **Power calculations must account for model choice**: Power under carryover misspecification cannot be extrapolated from no-carryover scenarios.

7.3 Implications for Trial Design

These findings have direct implications for precision medicine trial design:

- **OL+BDC designs** require carryover modeling to achieve adequate power
- **N-of-1 designs** with multiple crossovers are particularly vulnerable to carryover effects (not simulated in this paper)
- **Washout periods** of 5× the expected half-life may be insufficient for behavioral carryover, which a washout does not remove (Section 2.2); analytical approaches are preferred

Understanding this tradeoff enables more informed trial design decisions and more accurate power calculations for precision medicine validation studies.

Appendix: Generated Figures

The following figures were generated by `simulation_carryover_spectrum.R` and are available in the archived project at `archive/pmsimstats2025-2026-04-23/pmsimstats2025/analysis` (not reproduced here):

1. `carryover_spectrum_power.png`: power curves across the carryover spectrum
2. `carryover_spectrum_bias.png`: bias curves across the carryover spectrum
3. `carryover_spectrum_tradeoff.png`: combined bias-power visualization

References

Note (2026-10-07). All entries were checked in PubMed. Three were wrongly attributed in the original and are corrected: the Hendrickson et al. (2020) entry gave a nonexistent *Journal of Neurotrauma* article, the crossover-design review is by Lim and In (2021), not Lee and Ahn (2022), and the behavioral carryover paper is by Shi and Ye (2024) in *Biometrics*, not Zhou et al. in *Biostatistics*.

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